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[Signature]

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks.
(BL2, BL3)

Assessment Tool Weightage: 30%

1X100 = 100M

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships 20	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues 20	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships 10	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning 20	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map 15	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

85
100

88



Data Link vs Network Layer – Complete Data Delivery



Presented by: Pranav H (192525420)

CSA0720 Computer networks

How Data Link and Network Layers Ensure Complete Data Delivery

Data Link Layer (Layer 2)



Provides node-to-node delivery on the same network

MAC Address

00-1A-2B-3C-4D-5E



Unique hardware address assigned to each device

Local Delivery (Same Network)



Frames are delivered to the correct device within the local network using MAC addresses

Network Layer (Layer 3)



Provides end-to-end delivery across multiple networks

IP Address



Logical address assigned to each device for routing

Global Delivery (Across Networks)



Packets are routed across multiple networks to reach the destination using IP addresses



Complete End-to-End Data Delivery

1 Data Link Layer (Layer 2)



- Uses MAC addresses.
- Delivers frames within the same LAN.
- Performs framing and error detection (CRC).
- Ensures reliable local communication.

2 Network Layer (Layer 3)

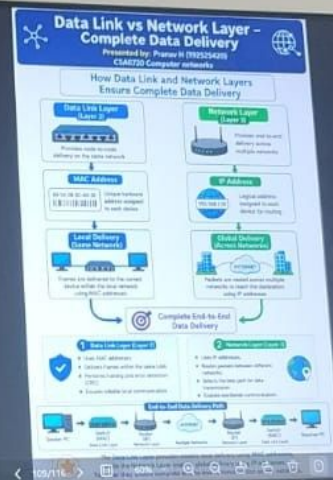


- Uses IP addresses.
- Routes packets between different networks.
- Selects the best path for data transmission.
- Enables worldwide communication.

End-to-End Data Delivery Path



The Data Link Layer provides reliable local delivery using MAC addresses, while the Network Layer enables global delivery using IP addresses. Together they ensure complete end-to-end communication across networks.



GPS Map Camera



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